Game-Theoretic Lane-Changing Decision Making and Payoff Learning for Autonomous Vehicles

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***Abstract*—In this paper, the problem of decision making for autonomous vehicles changing lanes is addressed by formulating multiple games in normal form for pairs of agents. This formulation generates the optimal action for the Ego vehicle at a given state and does not consider global optimality for all agents. The payoff matrices of the games are designed based on a user-defined set of rules. The constant parameters of these payoffs are then adjusted using neural learning to generate optimal behavior among the vehicles. An algorithm integrating deep reinforcement learning and game theory, regarded as Nash Q-learning, is included in the decision-making scheme. The applicability of the proposed method in a lane-changing scenario is tested via simulation.**

***Index Terms*—Autonomous driving, game theory, Nash equilibrium, neural networks.**

I. Introduction

In RECENT years, extensive research has been performed to improve the decision making capabilities of autonomous vehicles. Intelligent approaches have been particularly successful in solving complex autonomous driving and traffic control problems, with particular emphasis in the application of reinforcement learning algorithms [1]–[10]. Most of these results in the literature, however, treat the vehicle as a reactive system that executes actions as a direct response to its sensor inputs. This formulation of the problem neglects the fact that the road on which a vehicle travels is shared by other decision-making agents with their own interests and priorities.

Game theory is the field of mathematics that studies optimal decision making in multiagent systems [11], [12]. A game theoretic formulation allows to express both the cooperative and conflicting objectives of a group of agents in a formal mathematical framework that allows to determine their best responses in a given state. Game theory has been applied to different areas of traffic control [13]–[23]. In [24] a survey about the different applications of games to autonomous driving is presented. Common applications of these games include analysis of traffic fluidity, traffic signal coordination, ramp mettering control and vehicle-human interaction. In [13], [25]–[30], games between multiple vehicles are proposed to determine their best strategy in highway merging or lane-changing scenarios. Relevant variables that relate the states of the players at a given instant are defined and employed as game parameters. The solution of this games provides the optimal behavior for the vehicles. Here, we use a similar structure as starting point in our game theoretic formulation.

It is of our interest to design a decision making algorithm for a vehicle traveling on the road. A significantly useful tool in autonomous driving research would be the formulation of a game theoretic framework that allows the vehicles to determine when it is optimal to change lanes, to cross an intersection, to yield to other vehicles or pedestrians, and so on. In particular, this paper uses as a case of study a highway scenario on which the Ego vehicle must decide if it is safe to change lanes. In this scenario, Ego may have to compete against a rear vehicle in the target lane to occupy the available space, while at the same time having the common, cooperative goal of avoiding collisions.

In this paper, we consider a vehicle (Ego) that desires to change lanes in a highway, and formulate its competition against neighboring vehicles as a set of games in normal form. Each game is designed to consider both the conflicting goals (gain the space on a lane) and the cooperative goals (avoid collisions) of the vehicles. The solution of the game yields the optimal action for Ego at each particular state. To properly design the game payoffs, a predefined set of rules that describe a general appropriate behavior on the road is defined. Then, the parameters of these payoffs are tuned to yield the desired behavior using neural network training.

In the proposed decision-making strategy, the vehicles are not assumed to communicate their intentions with each other. The Ego vehicle only uses its sensor measurements to gather the state information of its neighbors, and then solves the formulated game to determine its optimal behavior. Since the neighboring vehicles may not behave optimally, we allow Ego to revise its decision at every time interval by re-playing the game using

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newly gathered data. Since we desire for these games to be played repeatedly, computational complexity considerations are taken into account in our algorithm design.

The contributions of this paper are the following. We propose a novel decision-making strategy for autonomous vehicles using a game-theoretic scheme. Different from current game-theoretic results in the literature, we design a fast decision-making algorithm that allows Ego to reconsider its decision at every time period using newly available data. This is achieved by formulating games between pairs of agents and therefore avoiding the computationally expensive solution of an N-dimensional game. We provide a method to determine the optimal decision for Ego from the solutions of multiple games. The result is a decision-making scheme that is safe, practical and scalable. Furthermore, we integrate for the first time a Nash Q-learning algorithm [31] in an autonomous vehicle decision-making strategy. This algorithm incorporates game theory, reinforcement learning and neural networks to determine the payoffs of the game that provide Ego with an optimal behavior. Finally, we present mathematical proofs about convergence and existence of solutions of the presented algorithm.

The presented contributions relate to the existing literature as follows. Many authors have studied the application of reinforcement learning approaches to self-driving vehicles (see [9] and references therein). In [10], for example, dynamic programming algorithms are designed for the optimization of energy consumption in electric vehicle applications. However, most of these publications do not consider a game-theoretical analysis of several decision-making agents as we do in this paper. In other publications, a game-theoretical approach is considered. In [29], [30], [32], Stackelberg games are formulated to determine the optimal actions of the vehicles. In these games, one of the vehicles acts as the leader while the other selects its action as a follower. Although Stackelberg games successfully model some of the vehicles interactions, they do not capture the dynamics of simultaneous decisions by the agents. Moreover, extending the results of these papers to multi-vehicle games leads to a considerable increase in computational complexity. In contrast, in this paper we allow Ego to aim for a Nash equilibrium solution given the current states of the interacting vehicles. Furthermore, our proposed method is designed to reduce the computational burden for the car's computer. A different approach, based in Bayesian games, was studied in [27], [28]. These papers consider the effect that the signals emitted by a vehicle have on the *beliefs* of the receiver. However, the solution of these games depends on the vehicles being able of interpreting the received signals. In our work, Ego only uses the state information measured with its sensors to determine its optimal actions, without any need to send or interpret particular information. Finally, in [13], [25], [26], game formulations similar to the one presented in this paper are proposed. These works use a direct rule-based approach for the selection of payoffs for the players. In contrast, we present and analyze a neural network method integrated with a Nash Q-learning algorithm to properly tune the game payoffs.

The structure of this paper is as follows. Section II introduces the basic definitions of game theory. In Section III a game is formulated between the Ego vehicle and its neighboring cars. Considerations about the efficiency and the reliability of the solutions are taken. The game payoffs for the players are defined in Section IV, along with the set of rules the vehicles are desired to follow. These payoffs are tuned by means of a neural network-based algorithm in Section V. Finally, VI presents numerical examples that clarify the applicability of our procedure. The conclusions of the paper are given in Section VII.

		B	
		L	R
A	U	$a_{11},\, b_{11}$	$a_{12},\, b_{12}$
	D	$a_{21},\, b_{21}$	$a_{22},\, b_{22}$

Fig. 1. Game in normal form. Each agent has two possible actions and their payoffs depend on each selection.

II. Game Theory Preliminaries

Game theory is the branch of mathematics that studies optimal decision making in multiagent interactions. In a general *normal-form game*, each member of a set of N players selects an action to carry out. Depending on the combination of actions executed, a payoff is assigned to each player. Formally, a game in normal form is defined as a tuple $(\mathcal{N}, A, J)$ where $\mathcal{N}$ s the set of N players, $A = A_1 \times \cdots \times A_N$ with A_i a set of actions available to player i, and $J = J_1 \times \cdots \times J_N$ where $J_i : A \to \mathbb{R}$ is the action-dependent payoff function for player i [11]. The objective of a player in the game is to obtain the highest possible payoff by selecting an appropriate action while taking into account that all other players will also try to maximize their own gains.

A game is often expressed by means of a table, as shown in Fig. 1. In this example there are two players, named A and B, each with two different actions to perform. Player A can execute actions U or D. Player B can perform actions L or R. The table displays the payoffs a_{ij} and b_{ij} that players A and B, respectively, would receive for a particular combination of actions.

Increasing the number of actions available to each player corresponds to an increase in the number of rows or columns in the payoff table. To consider additional players in the game, the table becomes a multi-dimensional array.

Let the set of actions available for player i be expressed as $A_i = \{s_i^1, \ldots, s_i^{m_i}\}$, where s_i^j represents the jth action available to player i. When there is no risk of ambiguity, we use the notation s_i to denote the action played by i. The symbol s_{-i} is used to represent the set of actions taken by all other players except player i, i.e., $s_{-i} = \{s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_N\}$.

The payoff received by player i during a game is expressed as $J_i(s) = J_i(s_i, s_{-i})$. The best response of player i for a given action profile s_{-i} is defined as the action s_i^* such that $J_i(s_i^*, s_{-i}) \geq J_i(s_i, s_{-i})$ for all actions s_i. Nash equilibrium (NE) is said to be achieved in the game if all players play their best response against each other, that is, all players select the

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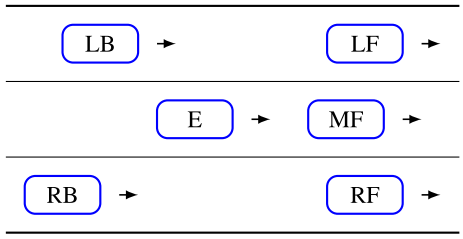

Fig. 2. Lane-changing scenario in a multi-lane highway.

actions s_i^* such that the inequalities

$$J_i(s_i^*, s_{-i}^*) \geq J_i(s_i, s_{-i}^*) \tag{1}$$

hold for $i = 1, \ldots, N$.

NE is the most important solution concept in game theory. This is a stable solution in the sense that, when the players are in NE, no player can unilaterally change its decision without negatively affecting their own payoff [11].

For some games, the Nash solution is given by a *pure strategy* by the players, that is, the optimal behavior of a player is to play a specific action with probability one. In general, this is not true for games in normal form. However, it has been proven [11] that in normal form games there always exist Nash solutions given by *mixed strategies*, where the optimal behavior of an agent is to select an action with a given probability.

III. Game Formulation for Highway Autonomous Driving

In this section, we describe a decision making scenario of an autonomous vehicle that desires to determine whether it is safe to change lanes while driving on a highway. The problem is formulated as a game in normal form as described in Section II. Practical considerations about decision reliability and computational complexity are taken into account.

A. Lane-Changing State Variables and Goals

Consider the scenario depicted in Fig. 2. The Ego vehicle (E) travels on a multi-lane highway shared by multiple other vehicles. We denote the relevant agents surrounding Ego as the left-rear (or left-back, LB), left-front (LF), right-rear (RB), right-front (RF) and middle-front (MF) vehicles. Fig. 2 shows a portion of the highway with three lanes going in the same direction, from left to right. In this scenario we have not considered a vehicle directly behind Ego (MB), as such vehicle would not affect Ego's decision to change lanes in our simplified formulation. However, the state of a MB vehicle may be necessary, for example, to determine if a lane-change maneuver can be aborted. These interactions will be considered in future versions of this research.

The state variables in this traffic scenario are defined as the longitudinal distances, d, and the velocities, v, of each vehicle with respect to Ego. The notation d_i is used to denote the longitudinal distance between Ego and vehicle i, where $i \in \{MF, LB, LF, RB, RF\}$. Similarly, v_i represents the velocity of vehicle i with $i \in \{E, MF, LB, LF, RB, RF\}$. Notice that v_E is defined but d_E is not. Given that we model the traffic scenario with Ego and its 5 surrounding vehicles, and that we exclude d_E as state variable, the total number of variables is 11. The state of each vehicle surrounding Ego is described by

$$x_i = \begin{bmatrix} d_i \\ v_i \end{bmatrix}, \tag{2}$$

$i \in \{MF, LB, LF, RB, RF\}$, while

$$x_E = v_E \tag{3}$$

In this formulation of lane-change decision making, we do not consider the lateral distances or velocities of the vehicles as a variable of interest.

Each vehicle can now use the state variables (2) and (3) to define two main goals at every instant of its travel in the highway. These main objectives are stated as follows,

Vehicle Goals

- Keep a safe distance from all other vehicles.
- Maintain a desired speed.

To achieve these goals, Ego must take into account that the agents around it have similar objectives. That is, Ego is not the only agent making rational decisions on the road, as the other vehicles are also interested in optimizing their own performance. The main advantage of a game-theoretic formulation is that, by considering the objectives of the other vehicles, we allow Ego to react not only to the current state of its surrounding environment, but also to the decisions other agents will make if they are to act optimally.

From Fig. 2 it becomes clear that, for this particular scenario, there are three players making decisions, namely, Ego, LB and RB. The cars in front of Ego are unlikely to react to Ego's needs to change lanes. The set of players in the game is thus defined as $\mathcal{N} = \{E, LB, RB\}$.

In this simplified problem, we let Ego have three possible actions available as *keep current lane* (KCL), *change lane to left* (CLL) and *change lane to right* (CLR); thus, $A_E = \{KCL, CLL, CLR\}$. Similarly, we allow the LB vehicle to have available only the two actions *reduce the speed* (RS) and *keep the speed* (KS). Of course, LB could decide also to accelerate, but let us use KS to represent anything that does not imply LB yielding the way to Ego. This means that LB can decide to slow down to allow Ego to change lanes, or it can decide to force Ego to wait by occupying the available space in the lane. The RB vehicle has the same actions available as LB, such that $A_{LB} = A_{RB} = \{RS, KS\}$.

To complete our game formulation as in Table 1, we must only define the payoffs J_i for all players in the game. These payoffs are defined in the next section. First, we propose a method to guarantee the reliability of the game-theoretic decision by continually replaying the formulated game.

B. Repeated Play With Changing Payoffs

By properly designing the game payoffs, the solution of a game generates an optimal decision for Ego at every given state. However, the states in a highway-driving game are subject to sudden changes such that the best action at a given instant may

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LB

		RS	KS
Ego	KCL	a^1_{11}, b^1_{11}	a^1_{12}, b^1_{12}
	CLL	a^1_{21}, b^1_{21}	a^1_{22}, b^1_{22}

Fig. 3. Game 1. Ego plays against LB only.

RB

		RS	KS
Ego	KCL	a^2_{11}, b^2_{11}	a^2_{12}, b^2_{12}
	CLR	a^2_{21}, b^2_{21}	a^2_{22}, b^2_{22}

Fig. 4. Game 2. Ego plays against RB only.

not be adequate a time period later. For example, the agents surrounding Ego may perform irrational actions, or unexpected environmental factors may emerge. Here a decision-making algorithm that remains safe against unforeseen circumstances is designed.

To avoid executing unsafe maneuvers, we propose to let Ego play the game repeatedly every time period. Repeating the game every few milliseconds allows Ego to quickly reconsider a taken decision according to the most recently acquired data. Consider, for example, the case on which an initial game yields the optimal decision for Ego to change lanes to the left. Soon after, however, LB suddenly increases its speed making Ego's decision unsafe. Then, at the next time period, Ego replays the game and decides that the optimal action is rather to keep its original lane. The lane-changing maneuver is aborted in a few milliseconds and Ego can perform a corrective action remaining safely in its current lane.

The repeated play procedure introduces an important practical constraint in the decision making process. Ego is required to compute the solution of a game in a fast and efficient manner before the next time period. This consideration is addressed in the following subsection.

C. Pairwise Games Formulation

The car velocities employed while traveling on a highway are often so high that the vehicle is desired to have a capacity of reaction of a fraction of a second. Every design step taken in favor of reducing the time of completion of the decision making procedure, without compromising the reliability of the solution, must be considered.

The three-player game described above implies the use of a three-dimensional array as payoff table. As the number of players in a game increases, the computational complexity required to solve it increases as well. Notice, however, that the lane-changing scenario in Fig. 2 allows us to solve this issue by decomposing the problem into pairwise games. Thus, let Ego play games against each of its neighboring vehicles separately.

In our highway scenario, Ego plays the first game against LB alone to decide whether it can change lanes to the left. The corresponding payoff matrix for this game is shown in Fig. 3. Notice that in this case Ego has only two actions available because here a change to the right lane is irrelevant. Concurrently, Ego plays a second game against RB to consider a right lane change, as shown in Fig. 4. The superindices in a^k_{ij} and b^k_{ij} indicate the game being played.

Each of these two-player game tables have only four cells, allowing a fast solution of the game by comparing the payoff values. The final decision of Ego about its lane of preference is determined from the solution to these games by means of the following rules.

While maintaining its current lane, the Ego vehicle plays Game 1 and Game 2 concurrently. If the solution of both games requires Ego to keep its lane, then that is Ego's final decision. If one and only one of the games has as solution a change of lanes, then Ego changes lanes to the corresponding side. If both games allow Ego to change lanes, then Ego changes lanes to the side that provides the higher payoff.

In the following section, the payoffs a_{ij} and b_{ij} for both games are designed using a set of common-sense rules that adequately represents the preferences of the cars to take a particular decision

IV. Payoff Design

In this section, we complete our game formulation to generate the optimal behavior for Ego. The state variables of the vehicles have been defined in (2) and (3) as the velocities and distances of the vehicles with respect to Ego. From this state information, it is possible to compute a suitable decision making variable to be used in the game payoffs. In the following subsection, the time-to-collision (TTC) between the vehicles is defined as an output of the controlled system.

A. Times-to-Collision as Decision Variables

The TTC between Ego and any other vehicle i is represented by T^i_C, and is only defined when the vehicle in the rear is faster than the vehicle in front. For example, if $v_E > v_{RF}$, then T^{RF}_C is defined as

$$T^{RF}_C = \frac{d_{RF}}{v_E - v_{RF}}. \tag{4}$$

Similarly, if $v_{LB} > v_E$, then T^{LB}_C is

$$T^{LB}_C = \frac{d_{LB}}{v_{LB} - v_E}. \tag{5}$$

In this sense, the TTCs are a state-dependent output of the system. If we let x_i be as in (2) and $y_i = T^i_C$, then

$$y_i = h(x_i) \tag{6}$$

for all $i \in \{MF, LB, LF, RB, RF\}$, where $h(x_i)$ is the corresponding function as in (4) or (5).

Remark 1: Notice that we have not set the requirement that the vehicles are in the same lane to define the TTC. We define the TTC as the time it would take the vehicles to collide if they were in the same lane. This definition allows the vehicles to determine the consequences of their lane-changing decisions.

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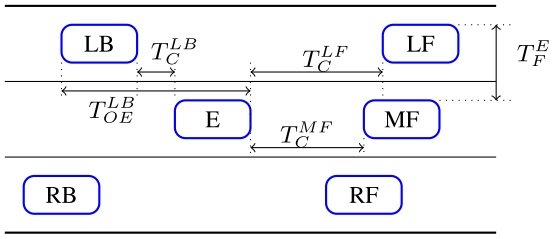

Fig. 5. Definition of variables for the agents payoff.

Fig. 5 schematically shows how the time to collision between Ego and other cars in the center and left lane are defined. Similar definitions are made for the right-lane vehicles, and they are omitted in the figure to improve clarity. The TTC defined as an output directly related to the position and velocities of the vehicles is a useful variable in the formulation of our game.

Besides the TTC, define additional useful variables. T_F^E is the time it takes for Ego to complete a lane-changing maneuver to either left or right lane. Notice that this variable implicitly accounts for the lateral distance Ego must travel to safely change its lane. This permits Ego to consider both longitudinal and lateral distances to finally make a decision. T_{OE}^{LB} and T_{OE}^{RB} represent the time to overtake Ego by LB and RB, respectively. Finally, the *imminent* time-to-collision (ITTC) between Ego and vehicle i is represented by T_{IC}^i and is defined as the value the TTC would have if Ego was moving at its desired speed, $v_{E,des}$. Thus, for example,

$$T_{IC}^{MF} = \frac{d_{RF}}{v_{E,des} - v_{MF}}. \tag{7}$$

It is of our interest to define the payoffs a_{ij} and b_{ij} in the Tables 3 and 4 using the variables here defined, such that the output of a vehicle at a given instant generates its optimal action. To determine appropriate expressions for these payoffs, we base our design in a set of decision rules, as described in the following subsection.

B. Output-Based Decision Rules

A set of simple rules is here defined to describe the expected behavior of each vehicle for a given state. Then, we use these rules to determine the game payoffs to generate such behavior.

Due to the symmetry between the games 1 and 2 in Figs. 3 and 4, we can focus on the design of one of the payoff tables and use similar results to design the other. Unless otherwise stated, in the remainder of this paper we study the game between Ego and LB only. Conclusions about the game against RB are then obtained.

Using the variables defined in the preceding subsection, let Ego consider the following rules to determine when to change lanes to the left. All $\bar{\theta}_j$ in these rules are constant parameters that will be determined in the next section.

1) If $\bar{\theta}_1 v_{MF} < v_{E,des}$, then change lanes.
2) If $\bar{\theta}_2 T_F^E > T_C^{LB}$, then keep current lane.
3) If $T_{IC}^{MF} < \bar{\theta}_3$, then change lanes.
4) If $\bar{\theta}_4 v_{LF} < v_{MF}$, then keep current lane.

The first rule says that Ego is encouraged to change lanes if its desired speed is higher than the current speed of the vehicle in front of it. From the second rule, Ego prefers to keep its lane if the TTC with respect to LB is smaller than the time it would take to complete the maneuver. The third rule expresses Ego's need to change lanes as the TTC with respect to MF becomes small. The fourth rule states that a lane change does not make sense if LF is slower than MF.

Similarly, define the rules of behavior for LB as follows,

- If $T_{OE}^{LB} < \bar{\theta}_5$, then keep the speed.
- If $v_E \approx v_{LB}$, then reduce the speed.

In this case, LB feels encouraged to keep its speed if it would not take long to pass Ego. LB considers a reduction of speed if it does not imply a significant change of speed.

Notice that as long as the rules are expressed in terms of both velocities and TTCs, they are inadequate to be compared with each other. To properly design the game payoffs using these rules, it would be convenient to express them in the same units. Consider, therefore, the following rule modifications,

1) Replace the rule $\bar{\theta}_4 v_{LF} < v_{MF}$ by $\bar{\theta}_6 T_C^{LF} < T_C^{MF}$.
2) Notice that the rule $T_{IC}^{MF} < \bar{\theta}_3$ already implies the rule $\bar{\theta}_1 v_{MF} < v_{E,des}$.
3) Replace the rule $v_E \approx v_{LB}$ by $T_C^{LB} > \bar{\theta}_7$.

From these rules, we are ready to define the game payoffs for both Ego and LB, as described in the following subsection.

C. Payoffs for Lane-Changing Games

Consider the payoffs a_{ij}^1 and b_{ij}^1 in Fig. 3. As we are working exclusively with the game between Ego and LB, we drop the superindex from this notation without risk of confusion.

From the rules defined above, Ego should keep its lane if

$$\bar{\theta}_2 T_F^E > T_C^{LB} \tag{8}$$

and

$$\bar{\theta}_6 T_C^{LF} < T_C^{MF}. \tag{9}$$

Using these rules, and reorganizing the constant parameters, define

$$a_{12} = \theta_1 (T_{IC}^{MF} - \theta_2 T_C^{LF}) + \theta_3 (\theta_4 T_F^E - T_C^{LB}). \tag{10}$$

Notice that the parameters θ_j are different from $\bar{\theta}_j$ for the convenience of renumbering. Now, as the inequalities (8) and (9) hold stronger, a_{12} in (10) becomes a larger positive number. From Fig. 3, this implies a higher payoff for Ego if it selects KCL as its action.

Similarly, Ego is encouraged to change its lane if $T_{IC}^{MF} < \bar{\theta}_3$. Therefore, we define

$$a_{21} = a_{22} = \theta_5 - \theta_6 T_{IC}^{MF}. \tag{11}$$

To define LB's payoffs, the rule $T_{OE}^{LB} < \bar{\theta}_5$ generates the expression

$$b_{21} = T_{OE}^{LB} - \theta_7 \tag{12}$$

and the rule $T_C^{LB} > \bar{\theta}_7$ directly implies

$$b_{12} = b_{22} = T_C^{LB} - \theta_8 \tag{13}$$

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Finally, notice that KCL/RS is an undesired outcome of the game under any circumstances. It is never optimal for the LB car to reduce its speed while Ego maintains its lane. Therefore, we define

$$a_{11} = a_{21} - \varepsilon \tag{14}$$

and

$$b_{11} = b_{12} - \varepsilon. \tag{15}$$

where ε is any positive constant. This selection of a_{11} and b_{11} prevents KCL/RS from being a Nash equilibrium of the game. Moreover, this selection has the significant consequence that it guarantees the existence of pure-strategy Nash equilibria (PSNE) in the game, as shown in the following theorem.

Theorem 1: Consider a two-player game where each player has two possible actions available. Let the game payoffs a_{11} and b_{11} be as in (14) and (15). Then, the game has at least one PSNE.

Proof: Notice that if (14) and (15) hold, then $a_{21} \geq a_{11}$ and $b_{12} \geq b_{11}$. Consider now the following two inequalities,

$$b_{21} \geq b_{22}, \tag{16}$$

$$a_{12} \geq a_{22}. \tag{17}$$

Now, if (16) is true and (17) is not, CLL/RS is the PSNE. If (17) is true and (16) is not, then KCL/KS is the PSNE. If both (16) and (17) are false, CLL/KS is the PSNE. Finally, if both (16) and (17) are true, then both CLL/RS and KCL/KS are PSNE in the game. □

The result in Theorem 1 fulfills an important requirement in our model. PSNE solutions provide the vehicle with a decision that must be carried out with probability one. On the other hand, if PSNE did not exist in the game, the vehicle would have to use a mixed strategy, and select an action with a given probability [11] making this procedure unsuitable for self-driving vehicles. Theorem 1 has also a fundamental role in Section V when we prove the convergence of the learning algorithm.

The final decision of the players in the game depends on the values of the parameters θ_j that are still to be defined. In the following section, we propose a neural network (NN) algorithm to determine the values of these constants.

V. Payoff Learning Algorithm

Consider the neural network structure shown in Fig. 6. This is a neural network with TTCs as inputs, game payoffs as hidden-layer outcomes and Nash decisions as outputs. The objective of this network is to update the values of the parameters θ for the game payoffs of Fig. 3. These parameters θ are defined as the neural weights of the first NN layer. The connections of the neurons are designed such that the output of the first layer is the value of the payoffs a_{ij}, b_{ij} as defined in Section III.

The NN output layer then compares the payoffs in order to solve the game and determine the best actions for each vehicle. The weights of the links between the hidden layer and the output layer are static, such that the same operations are always performed.

By means of the functions ϕ_{ij}, the output layer compares the payoffs to determine the Nash solution of the game. Select, for

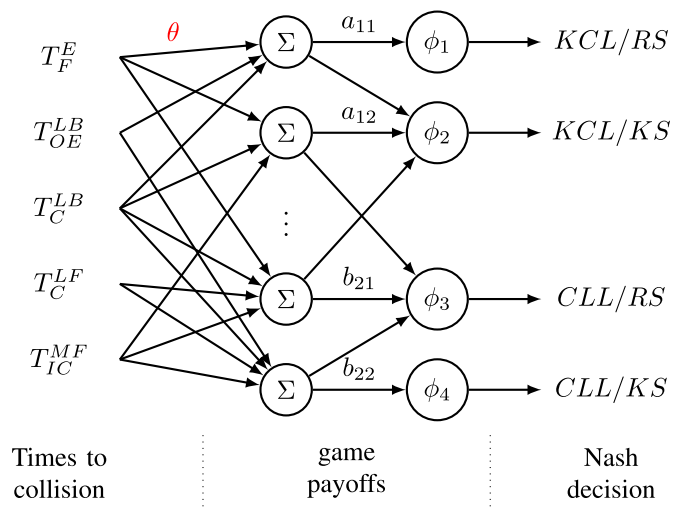

Fig. 6. Neural network for payoff learning.

example, the functions

$$\phi_{11} = \tanh(\beta(a_{11} - a_{21})) + \tanh(\beta(b_{11} - b_{12}))$$
$$\phi_{12} = \tanh(\beta(a_{12} - a_{22})) + \tanh(\beta(b_{12} - b_{11}))$$
$$\phi_{21} = \tanh(\beta(a_{21} - a_{11})) + \tanh(\beta(b_{21} - b_{22}))$$
$$\phi_{22} = \tanh(\beta(a_{22} - a_{12})) + \tanh(\beta(b_{22} - b_{21}))$$

such that $\phi_{22} \approx 2$ if CLL/KS is the Nash equilibrium.

Fig. 6 aims to describe the learning method on which the first layer of the network computes the payoffs a_{ij} and b_{ij}, and the output layer compares these values to determine the Nash solution of the game.

For this neural learning formulation to work it is necessary that there exist weights θ that allow an appropriate decision making by a vehicle. In the following theorem, it is shown that such weights exist as long as the decision variables for both possible decisions of a vehicle are linearly separable.

Theorem 2: There exist parameters θ that provide the optimal lane-changing decision for Ego if the neural input values that must yield the action CLL are linearly separable from the neural input values that must yield KCL. Similarly, there exist parameters θ that provide the optimal action for LB if the neural input values for each decision are linearly separable.

Proof: By definition, two sets S and T are linearly separable if there exist scalars ρ_i, $i = 1, \ldots, n$, and κ, such that for every $s \in S$,

$$\sum_{i=1}^{n} \rho_i s_i > \kappa$$

and for every $t \in T$,

$$\sum_{i=1}^{n} \rho_i t_i < \kappa$$

Now, for KCL to be the Nash equilibrium decision for Ego, it is necessary that $a_{12} > a_{22}$; that is,

$$\theta_1(T_{IC}^{MF} - \theta_2 T_C^{LF}) + \theta_3(\theta_4 T_F^E - T_C^{LB}) > \theta_5 - \theta_6 T_{IC}^{MF}$$

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Reorganizing this inequality, we get

$$(\theta_1 + \theta_6)T_{IC}^{MF} - \theta_1\theta_2 T_C^{LF} + \theta_3\theta_4 T_F^E - \theta_3 T_C^{LB} > \theta_5$$

Thus, if the neural input vector $[T_F^E, T_C^{LF}, T_{IC}^{MF}, T_C^{LB}, T_{OE}^{LB}]^T$ is linearly separable, there exist suitable parameters θ to generate this solution.

A similar proof shows that there exist parameters θ for LB's optimal decision, given that its best decision is KS if the inequality

$$T_C^{LB} - \theta_8 > T_{OE}^{LB} - \theta_7$$

holds. ■

Obtaining the training data set for the NN can be achieved using two possible approaches. First, we could collect data from real-time applications on which an expert driver executes the best actions at each state. However, this approach is limited by the number of different scenarios that the driver would encounter on the road. Moreover, the actions performed during the collection of data are prone to human error, and there is no guarantee that these actions are truly optimal.

The second option is to design an unsupervised algorithm that determines the optimal actions at every state by exploring the set of possible state-action combinations. This learning scheme leads us to the formulation of a Q-learning algorithm, as described in the next subsection.

A. Nash Q-Learning for Autonomous Driving

The game formulation here presented introduces two decision making agents, each seeking to maximize its payoff. The conventional Q-learning algorithm must then be be modified to encompass the combined effect of the actions of both agents. In [31] a Nash Q-learning algorithm is designed to learn the payoffs of a game in normal form; conditions for the convergence of the algorithm are there provided. In this section we apply this procedure to determine the optimal action of the vehicles at a given state in the proposed game formulation.

1) Rewards: First, it is necessary to design high-level rewards for the vehicles. These rewards must measure the quality of an action with respect to the vehicle goals presented in Section III. Consider the Ego vehicle and its goal of maintaining a desired speed during its travel in this section of the highway. A suitable reward for its actions may be defined as

$$r_{E,v} = e^{-0.1(v_E - v_{E,des})^2} \tag{18}$$

where $v_{E,des}$ is the desired velocity. On the other hand, the goal of maintaining a safe distance from the other vehicles can be expressed by the reward

$$r_{E,d} = \begin{cases} e^{-0.1(d_{min} - d_{safe})^2/d_{safe}} & \text{if } d_{min} < d_{safe} \\ 0 & \text{otherwise} \end{cases} \tag{19}$$

where d_{min} is the minimum longitudinal distance between Ego and the other vehicles in its lane, i.e., the distance with respect to its closest neighbor. d_{safe} is the safe distance that Ego desires to maintain with respect to every other vehicle. Thus, the reward (19) penalizes Ego when it gets closer to any vehicle in its lane than a predefined safe distance. Fig. 7 displays the shape of the functions $r_{E,v}$ and $r_{E,d}$.

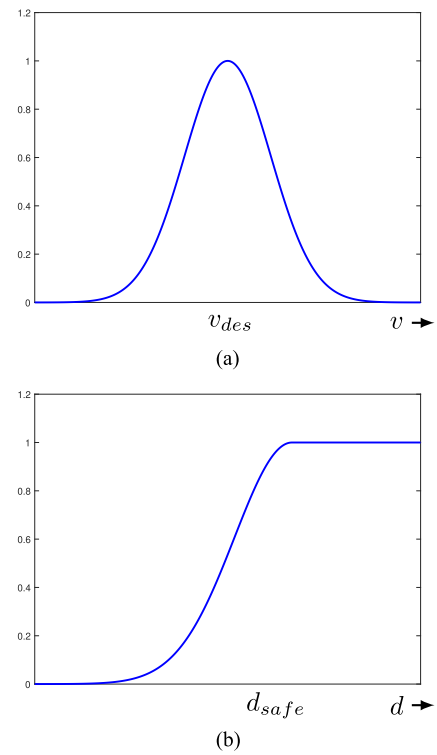

Fig. 7. a) Velocity reward function. b) Distance reward function.

Define the final reward for Ego as

$$r_E = c_E r_{E,v} + (1 - c_E) r_{E,d} \tag{20}$$

where $0 < c_E < 1$ is a constant parameter that determines Ego's priority for increasing $r_{E,v}$ or $r_{E,d}$. The larger the value of c_E, the more preference is given to $r_{E,v}$, which produces a more aggressive behavior by Ego on the road. Using similar expressions, we define the reward for LB, r_{LB}.

Remark 2: The selection of the rewards for different applications is not unique, and different alternatives can be considered depending on the desired performance. Notice, however, that many different driving goals can be expressed in terms of the two main objectives here selected for the vehicles. For example, the desire to minimize fuel consumption can be implicitly addressed by a proper design of the desired velocity.

2) Nash Q-Learning: Nash Q-learning is a method to determine the optimal action of an agent in a normal-form game. Its main difference from regular Q-learning is that, in game theory, the optimal solution is not necessarily the highest payoff, but the so-called Nash solution.

In the Nash Q-learning algorithm, the Q-value for Ego corresponding to taking action a_E when LB uses action a_{LB} at time t, is updated using the following formula

$$Q_E(x_t, a_{E,t}, a_{LB,t}) \leftarrow (1 - \alpha_t) Q_E(x_t, a_{E,t}, a_{LB,t}) + \alpha_t \left[r_{E,t+1} + \gamma \underset{a_E, a_{LB}}{Nash}\, Q_E(x_{t+1}, a_E, a_{LB}) \right] \tag{21}$$

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Notice that, instead of using the $\max$ operator as in conventional Q-learning, we make use of the Nash operator to determine the optimal actions for both players.

The corresponding update formula for Q_{LB} is given by

$$Q_{LB}(x_t, a_{E,t}, a_{LB,t}) \leftarrow (1-\alpha_t)Q_{LB}(x_t, a_{E,t}, a_{LB,t}) + \alpha_t \left[r_{LB,t+1} + \gamma \underset{a_E, a_{LB}}{Nash}\, Q_{LB}(x_{t+1}, a_E, a_{LB}) \right] \quad (22)$$

In [31], the proof of convergence of these update laws requires the technical condition that the Nash operator provides the highest payoffs for the agents in the game, or that they form a saddle point. This is a restrictive condition that our formulation avoids.

In the following theorem, it is proven that the conditions for convergence of the Nash Q-learning algorithm are relaxed for the game formulated in this paper. This is achieved by enforcing the conditions (14) and (15) in the update rule of the Q-values, since Theorem 1 shows the resulting existence of PSNE solutions in the game.

Theorem 3: Let the sequences $Q_{E,t}$ and $Q_{LB,t}$ be updated respectively by

$$Q_{E,t+1}(x_t, a_{E,t}, a_{LB,t}) = (1-\alpha_t)Q_{E,t}(x_t, a_{E,t}, a_{LB,t}) + \alpha_t \left[r_{E,t+1} + \gamma Q_E(x_{t+1}, \bar{a}_E, \bar{a}_{LB}) \right] \quad (23)$$

and

$$Q_{LB,t+1}(x_t, a_{E,t}, a_{LB,t}) = (1-\alpha_t)Q_{LB,t}(x_t, a_{E,t}, a_{LB,t}) + \alpha_t \left[r_{LB,t+1} + \gamma Q_L B(x_{t+1}, \bar{a}_E, \bar{a}_{LB}) \right], \quad (24)$$

where $(\bar{a}_E, \bar{a}_{LB})$ are the corresponding PSNE actions for agents E and LB, and define

$$Q_E(x, KCL, RS) = Q_E(x, CLL, RS) - \varepsilon, \quad (25)$$

and

$$Q_{LB}(x, KCL, RS) = Q_{LB}(x, KCL, KS) - \varepsilon, \quad (26)$$

for some $\varepsilon > 0$. Then, the sequences converge to the Nash Q-values Q_E^* and Q_{LB}^*.

Proof: By Theorem 1, conditions (25) and (26) guarantee the existence of a PSNE in the game. Therefore,

$$\underset{a_E, a_{LB}}{Nash}\, Q_E(x_t, a_E, a_{LB}) = Q_E(x_t, \bar{a}_E, \bar{a}_{LB})$$

and the updates (23) and (24) correspond to the updates (21) and (22).

We aim to prove that

$$P_t Q_E(x_t, a_{E,t}, a_{LB,t}) = r_{E,t+1} + \gamma Q_E(x_{t+1}, \bar{a}_E, \bar{a}_{LB})$$

is a contraction operator.

Consider any two Q functions Q_E and $\hat{Q}_E$. Then,

$$\begin{aligned} \|P_t Q_E - P_t \hat{Q}_E\| &= \max_x |P_t Q(x) - P_t \hat{Q}(x)| \\ &= \max_x |r_{t+1} + \gamma Q(x, \bar{a}_E, \bar{a}_{LB}) - r_{t+1} \\ &\quad - \gamma \hat{Q}(x, \bar{a}_E, \bar{a}_{LB})| \\ &= \max_x |\gamma Q(x, \bar{a}_E, \bar{a}_{LB}) - \gamma \hat{Q}(x, \bar{a}_E, \bar{a}_{LB})| \\ &= \gamma \max_x |Q(x, \bar{a}_E, \bar{a}_{LB}) - \hat{Q}(x, \bar{a}_E, \bar{a}_{LB})| \\ &\le \gamma \max_x \max_{a_E, a_{LB}} |Q(x, a_E, a_{LB}) \\ &\quad - \hat{Q}(x, a_E, a_{LB})| \\ &= \gamma \|Q - \hat{Q}\| \end{aligned}$$

This completes the proof. □

Algorithm 1: Nash Q-Learning for Autonomous Driving.

procedure
  Initiate the deep network $Q(w)$ with parameters w.
  for episode=1 to N_e,**do**
    Set random initial conditions for Ego, MF, LF and LB.
    for samples t=1 to N_s, **do**
      Select the Nash actions $\bar{a}_E$ and $\bar{a}_{LB}$. With probability ε, each car selects the opposite action.
      Apply actions a_t. Determine and store the rewards $r_{E,t+1}$ and $r_{LB,t+1}$.
      Set $z_{E,j} = r_{E,j+1} + \gamma \hat{Q}_E(x_{j+1}, \bar{a}_E, \bar{a}_{LB}, w)$, where $(\bar{a}_E, \bar{a}_{LB}) = \arg \underset{a_E, a_{LB}}{Nash}\, Q_E(x_{j+1}, a_E, a_{LB})$.
    end for
    Every N_r episodes, update the neural parameters w by performing gradient descent on $(z_j - Q(s, a, w))^2$
    Every N_c episodes set $\hat{Q} = Q$.
  end for
for procedure

The final Nash Q-learning algorithm is given in Algorithm 1. To obtain the current estimations of the Q-values, a deep NN is employed and updated every N_r iterations of the algorithm.

Remark 3: Algorithm 1 allows Ego to explore a wide range of possible driving scenarios, such that its best response is adapted to diverse driving conditions. This information can be used to train the NN that tunes the parameters θ described in Section IV-C. On the road, Ego takes advantage of this experience as it determines its actions according to its current conditions, and as it adapts its behavior when necessary by using the repeated play feature of our decision-making method.

The final decision making scheme is shown in Fig. 8. Our game formulation plays the role of an *actor*, indicating Ego its optimal decision for each particular output vector y. During the learning stages, this actor is updated by a *critic* that uses the state information of the vehicles to compute the corresponding rewards $r(x)$. In our formulation, this critic corresponds to both the Nash Q learning algorithm, followed by the neural network training to obtain the parameters θ.

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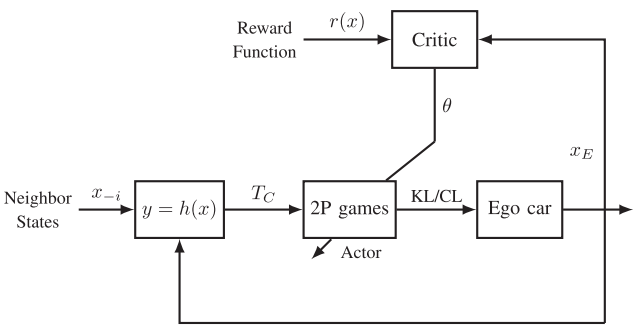

Fig. 8. Desicion-making scheme for highway driving.

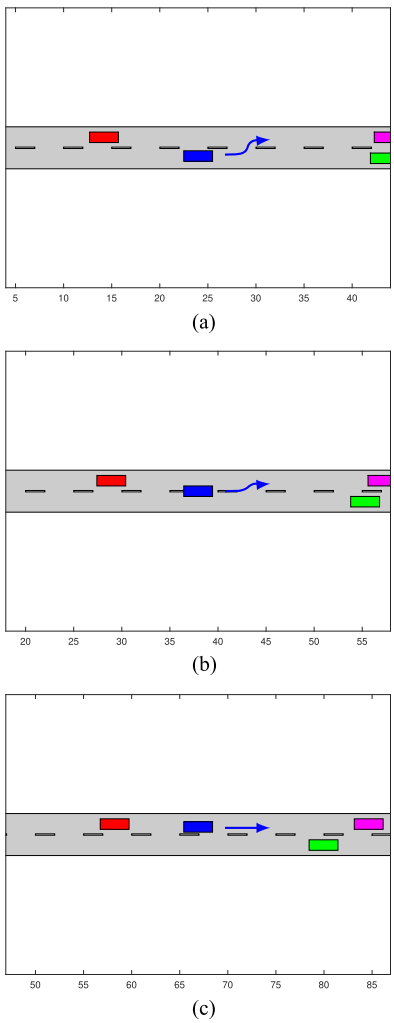

Fig. 9. Confirmed decision. a) Ego decides to change lanes. b) The maneuver is executed. c) Maneuver completed.

VI. Simulated Implementation Using Vehicle Dynamics

To test the proposed decision making procedure, a simulation of the lane-changing scenario is performed. In this simulation, the vehicles dynamics are obtained using the dynamic bicycle model [33]

$$\dot{v}_x = v_y\dot{\psi} + a,$$

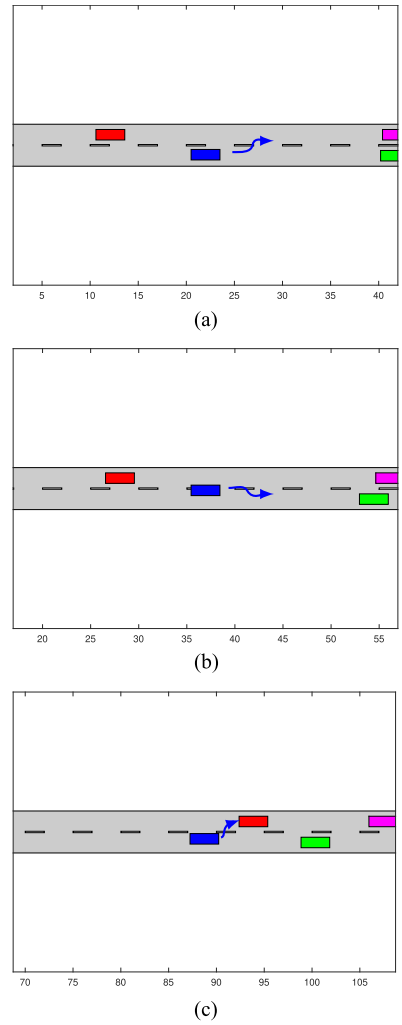

Fig. 10. Revised decision. a) Ego decides to change lanes. b) A sudden change in the speed of LB makes Ego change its decision. c) Ego returns to its original decision.

$$\dot{v}_y = -v_x\dot{\psi} + \frac{2}{m}\left(F_{cf}\cos(\delta_f) + F_{cr}\right),$$

$$\ddot{\psi} = \frac{2}{I_z}\left(l_f F_{cf} - l_r F_{cr}\right),$$

$$\dot{X} = v_x\cos(\psi) - v_y\sin(\psi),$$

$$\dot{Y} = v_x\sin(\psi) + v_y\cos(\psi),$$

where X and Y are the longitudinal and lateral positions, respectively, in the body frame; v_x and v_y are the longitudinal and lateral speeds of the center of mass, respectively, in an inertial frame; ψ is the yaw angle; m and I_z are the mass and yaw inertia, respectively; and F_{cf} and F_{cr} are the lateral tire forces at the front and rear wheels, respectively. F_{cf} and F_{cr} are computed as $F_{ci} = -C_{\alpha_i}\alpha_i$, for $i \in \{f, r\}$, where the tire cornering stiffness C_{α_i} and the tire slip angle α_i are used. The lengths from the center of mass to the front and rear ends of the car are here taken as $l_f = l_r = 1.5\tilde{}m$, such that the total car

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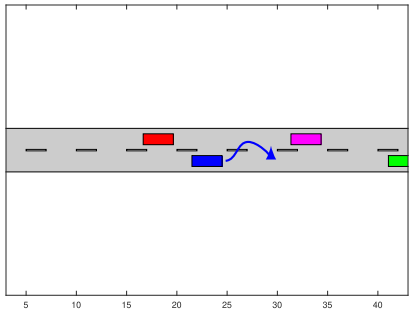
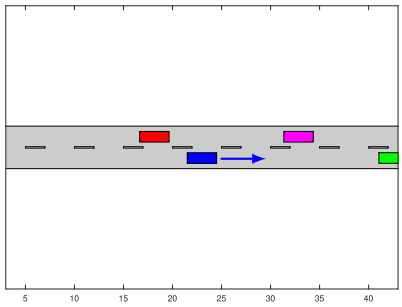


(a)



(b)

Fig. 11. Different decisions obtained from a regular Q-learning training and the game-theoretic approach proposed in this paper.

		LB	
		RS	KS
Ego	KCL	$-0.768,\ 0.409$	$0.458,\ 0.509$
	CLL	$-0.668,\ 0.383$	$-0.668,\ -0.509$

Fig. 12. Values of the payoff table when LB decreases its speed.

length is $L = l_f + l_r = 3\tilde{}m$. The inputs of the system are the acceleration a and the front steering angle δ_f.

A. Algorithm Implementation

The simulations here presented were carried out using the following information. Ego travels in a highway section surrounded by other vehicles. During its travel, we control the longitudinal velocity and the lateral position of Ego. Let Ego have as desired longitudinal velocity $v_{E,des} = 20\tilde{}m/s$. While the desired velocity remains constant, the *reference velocity*, $v_{E,ref}$ varies according to the traffic circumstances. The reference velocity is the target velocity for the controller, as described below. If there is no risk of collision, then Ego's reference velocity is equal to its desired velocity; if the vehicle in front, say MF, is slower than Ego, then Ego must match MF's speed for as long as it remains on the same lane. Thus, $v_{E,ref} = \max\{v_{E,des},\ v_{MF}\}$.

The reference for Ego's lateral position, $y_{E,ref}$, is provided by the solution of our game-theoretic algorithm. The starting reference position for Ego is its current lane. Then, if the game optimal decision is to change the lane, then $y_{E,ref}$ varies accordingly. If the decision is to keep the lane, $y_{E,ref}$ remains unchanged.

The low-level controller provides the selection of the inputs a and δ_f. The longitudinal velocity is controlled using a PID input [34], where the velocity error is defined as $e_v = v_{x,E} - v_{E,ref}$. We furthermore force the acceleration of the vehicles to be bounded by the range $[-3\tilde{}m/s,\ 3\tilde{}m/s]$. The lateral position control is achieved by using a Stanley controller [35] as

$$\delta_f = \tan^{-1}\left(\frac{2\tilde{}Le_y}{v_{x,E}}\right) + \psi$$

where the lateral position error is $e_y = Y_E - y_{E,ref}$.

To determine the payoff parameters θ, the NN in Fig. 6 is employed. Therefore, the first step is to use Algorithm 1 to generate a data set to train this network. The rewards employed in Algorithm 1 are selected as in (18) and (19), using the following parameters. The distance d_{min} is defined as the minimum distance between Ego and the other vehicles in its current lane. For example, if Ego changes lanes and finds itself between LF and LB, then $d_{min} = \min\{d_{LF},\ d_{LB}\}$. The safe distance d_{safe} that Ego desires to maintain is selected as $d_{safe} = 40\tilde{}m$. Finally, the outputs of the deep network $Q(w)$ is initiated as zeros for all agents and actions. After learning, this algorithm provides us with samples of the optimal action of each vehicle at different states.

Using this information, the NN in Fig. 6 is trained and the parameters θ are determined. These parameters are now used in the game payoffs (10)–(15) that determine the behavior of the vehicles during their travel in the highway.

Different lane-changing circumstances are now simulated. The vehicles surrounding Ego perform a variety of predefined actions that allow us to test the behavior of Ego. The following figures present some of these cases. First, Fig. 9 shows a usual scenario on which Ego has enough time to change lanes to left, successfully completing the maneuver.

A more interesting case is shown in Fig. 10. Here, to test the applicability of the game replay proposed in Section III, we let the LB vehicle (red) behave irrationally. Ego starts with enough time to change lanes and starts the maneuver. Then, LB suddenly accelerates making Ego's previous decision unsafe. Fig. 10 shows that Ego responds by changing its decision before completing the lane change, returning safely to its original lane. Ego changes lanes again when LB has passed.

B. Comparison With Regular Q-Learning

We complete the simulation tests of our decision-making method by comparing it with existing vehicle control algorithms. In particular, a deep Q-learning algorithm, without a game formulation, is compared to the Nash Q-learning algorithm presented in this paper. This comparison shows the advantages of a game-theoretical approach.

The same simulation settings described above are used for the deep Q-learning algorithm. This includes the vehicle characteristics, the low-level controller design and the reward selection. Instead of finding the Nash equilibrium, the Q-learning algorithm must decide when Ego can safely change lanes by

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maximizing the discounted return

$$R = \sum_{k=0}^{\infty} \gamma^k r_{E,k},$$

where $0 < \gamma \leq 1$ is the discount factor. Naturally, we explore in this example the limits of this learning algorithm and show that a better behavior can be obtained by application of game theory.

After learning is complete, let Ego react to a double change of speed in the LB vehicle. LB is programmed to maintain a speed of $21\,m/s$ for one second, then slow down to $20\,m/s$ as if to yield the way for Ego, and then speed up to $21\,m/s$ after one second. Fig. 11(a) shows the observed behavior of Ego. Notice that it initiates a lane-change maneuver when LB reduces its speed. This decision seems to make sense as Ego has enough time to change its lane. Then, as the LB vehicle speeds up, Ego must go back to its original lane.

Fig. 11(b) shows that Ego improves its behavior when using games to decide its action. In this case, Ego does not change lanes immediately after noticing the reduction of speed in LB. This is because, in this simulation, the payoff table at this instant is as shown in Fig. 12. The Nash solution of this game, despite the reduction of LB's speed, mandates Ego to keep its current lane (KCL) and LB to keep its speed (KS). Thus, by considering the goals of LB, Ego remains in its lane.

VII. Conclusion

A game-theoretic scheme was presented to achieve optimal decision-making in a lane-changing scenario. The proposed formulation is designed to be fast and reliable, two major concerns in any real-time implementation. The method is fast as Ego can effortlessly solve pair-wise games, and then integrate these solutions to make a decision. This greatly reduces the time of computation required to solve a multi-player, N-dimensional game. The method is reliable because Ego revises its decision at every time step, modifying its behavior accordingly. An intelligent algorithm was designed to tune in the game parameters and adjust the behavior of the players as desired. Simulation results show the advantages of the proposed decision-making strategy, as the vehicle considers other agents objectives before selecting its optimal action. Notice that this method provides a feasible solution to Ego-vehicle decision-making and does not solve a global optimal problem for all the agents.

This work is being continued considering game-theoretic formulations of alternative driving scenarios like intersection crossing. Further extensions of this work include complementing the current simplified scenario to include more complex traffic conditions and more complete action availability for the players of a game. Moreover, additional optimization variables, like fuel economy, are being considered in the optimization scheme.

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